Deepak Mallampati

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Kent, OH - Open to relocation to Paris (visa sponsorship required)

PROFESSIONAL SUMMARY

Applied AI engineer (2.5+ years) specializing in LLM evaluation infrastructure, grounding-based quality assessment, and production-readiness frameworks. Master's degree (Kent State University). Designed evaluation pipelines, grounding rules, hallucination tracking systems, and domain-specific recall metrics for production healthcare AI - retrieval-grounded alignment exceeded 90%, hallucination reduced >30%, recall improved 15-20% on high-risk categories. Strong Python depth with Hugging Face Transformers, PyTorch, and RAG/agentic system experience. Seeking first Evaluation Engineer role at Mistral AI to define what "production-ready" means at frontier model scale.

CORE COMPETENCIES

Evaluation Pipelines & Frameworks • LLM Grounding & Guardrails • Hallucination Tracking • Retrieval-Augmented Generation (RAG) • Agentic System Benchmarking • Domain-Specific Evaluation (Healthcare) • Hugging Face Transformers / PyTorch • LangChain / LlamaIndex • Python / FastAPI • Audit Trails & Compliance • Precision / Recall / Threshold Calibration • Production AI Delivery

PROFESSIONAL EXPERIENCE

Progress Solutions Inc. *Jul 2025 - Present*

Jr. AI/ML Engineer Trainee - Healthcare AI, USA

- Designed **evaluation framework** for production healthcare RAG: grounding rules, hallucination tracking, retrieval alignment metrics, and audit trails - retrieval-grounded alignment exceeded **90%**; hallucination reduced **>30%**.
- Built **domain-specific eval harness** for high-risk patient categories: class weighting, stratified sampling, threshold calibration by patient risk tier - recall improved **15-20%** on high-risk patients while holding precision **>90%**.
- Developed **agentic LLM eval system** measuring per-step grounding, tool discipline, and structured output fidelity across multi-step healthcare workflows - agent stability improved **~25%**.
- Improved contextual retrieval precision **~35%** via recursive semantic chunking and transformer-based embeddings; implemented LangChain and LlamaIndex retrieval pipelines.
- Packaged ML/LLM inference as **FastAPI/Docker** containerized services with structured logging - reduced post-deployment defects **~30%**.
- Maintained HIPAA-conscious data governance: de-identification, data lineage, evaluation audit trails, stakeholder system-limitation documentation.

Suvidha Foundation *Jan 2022 - Mar 2022*

Machine Learning Engineer - Nonprofit Tech, Hyderabad, India

- Designed **human-in-the-loop evaluation methodology** for transformer-based video summarization - achieved **~85% alignment** with human-curated key moments across 5,000+ sessions.
- Built **RAG document Q&A evaluation**: reduced hallucinated outputs **~30%**, raised contextual answer accuracy **>85%** in controlled evaluation framework.
- Deployed evaluation-validated systems as Flask API for non-technical staff - 60-70% reduction in manual review time.

Energy Solutions International (Emerson) *Jun 2022 - Apr 2023*

Database & DevOps Performance Engineer (Intern) - Hyderabad, India

- Built performance monitoring dashboards (SQL Server DMVs + Grafana) - systematic observability infrastructure reducing incident recurrence **~25%**.
- Built Jenkins CI/CD pipelines for SQL schema validation and deployment - reduced deployment errors **>30%**.

SELECTED PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Eval-first multi-agent system - **schema-validated JSON contracts between agents** as inter-agent evaluation checkpoints. RAG-grounded CPT/ICD validation against vector-indexed policy documents; audit-ready reasoning traces for each fraud risk scoring decision. Evaluation built into the architecture from day one.

LangChain, LlamaIndex, Python, FastAPI, Docker

Healthcare RAG Evaluation Pipeline

End-to-end evaluation infrastructure for clinical RAG: grounding rules engine, retrieval quality scoring, hallucination rate tracking by query category, threshold calibration framework, and audit-trail export for compliance review. Reproducible, automated, and production-grade.

Python, LangChain, LlamaIndex, Pandas, NumPy

Agentic Pixel Character Synthesis & Animation Engine

Generative AI system with LLM-based agent orchestrator - each stage has defined evaluation criteria (identity consistency, pose accuracy, animation smoothness). Stable Diffusion fine-tuning + LoRA + ControlNet + OpenPose/MediaPipe.

PyTorch, Hugging Face Diffusers, FastAPI, Docker, LLM orchestration

EDUCATION

Master's Degree - *Kent State University, Ohio, USA* *Aug 2023 - May 2025*

TECHNICAL SKILLS

Evaluation & Quality: Evaluation pipelines, grounding rules, guardrails, hallucination tracking, retrieval alignment metrics, audit trails, precision/recall calibration, human-in-the-loop eval

LLM & GenAI: LangChain, LlamaIndex, RAG, agentic workflows, structured outputs, prompt engineering, embeddings, vector search

ML Libraries: PyTorch, Hugging Face Transformers, scikit-learn, XGBoost, Pandas, NumPy

Languages & Infra: Python, FastAPI, Flask, SQL, Docker, Jenkins, Grafana, CI/CD, MLOps/LLMOps

Work Authorization: F-1 OPT (US-authorized); requires visa sponsorship for France/EU